The documentation and process conversion measures necessary to comply with this revision shall be completed by 25 January 1994

INCH-POUND

MIL-S-19500/181E
25 October 1993
SUPERSEDING
MIL-S-19500/181D
27 July 1983

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER
TYPES 2N718A, 2N1613 AND 2N1613L, JANTX AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN silicon, low-power transistors. Two levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. Type 2N718A, see figure 1 (TO-18), types 2N1613 and 2N1613L, see figure 3 (TO-39 and TO-5).

1.3 Maximum ratings. T_C = +25°C unless otherwise specified.

Type	P_T 1/ T_C = +25°C	P_T 2/ T_A = +25°C	V_{CBO}	V_{CEO}	V_{EBO}	I_C	V_{CER} R_{BE} = 10 Ω	$R_{\theta JC}$	T_J and T_{STG}
	W	W	V dc	V dc	V dc	A dc	V dc	°C/W	°C
2N718A	1.8	0.5	75	30	7	500	50	97	-65
2N1613	3.0	0.8	75	30	7	500	50	58	to
2N1613L	3.0	0.8	75	30	7	500	50	58	+175

1/ Derate linearly at 17.2 mW/°C for type 2N1613 and 2N1613L and at 10.3 mW/°C for type 2N718A for T_C > +25°C.

2/ Derate linearly at 4.57 mW/°C for type 2N1613 and 2N1613L and at 2.85 mW/°C for type 2N718A for T_A > +25°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defence Electronics Supply Center ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5270, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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1.4 Primary electrical characteristics.

Limits	h_{FE1} 1/	h_{FE2} 1/	h_{FE3} 1/	h_{FE4} 1/	$V_{CE(SAT)}$ 1/	$\lvert h_{fe} \rvert$ at 20 MHz
	V_{CE} = 10 V dc I_C = 0.1 mA dc	V_{CE} = 10 V dc I_C = 10 mA dc	V_{CE} = 10 V dc I_C = 150 mA dc	V_{CE} = 10 V dc I_C = 500 mA dc	I_C = 150 mA dc I_B = 15 mA dc	V_{CE} = 10 V dc I_C = 50 mA dc f = 20 MHz
					V dc	
Min	20	35	40	20		3
Max			120		1.5	

1/ Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

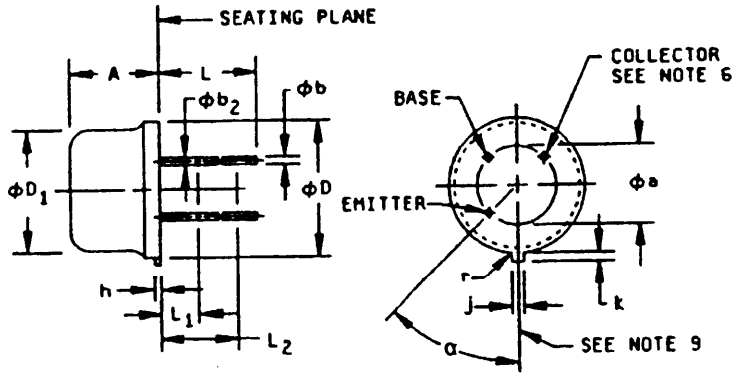
3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and on figures 1 and 3 herein.

3.3.1 Lead finish. Lead material shall be Kovar or Alloy 52. Lead finish shall be solderable in accordance with MIL-STD-750 and MIL-S-19500. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.170	.210	4.32	5.33	
φa	.707 Nom		1.796 Nom		5
φb		.021		.53	3, 8
φb2	.016	.019	.41	.48	4, 8
φD	.209	.230	5.31	5.84	
φD1	.178	.195	4.52	4.95	
h		.020		.51	
j	.036	.046	.91	1.17	
k	.028	.048	.71	1.22	7
L	.500		12.70		8
L_1		.050		1.27	8
L_2	.250		6.35		8
r		.007		.18	
α	45° T.P.		45° T.P.		

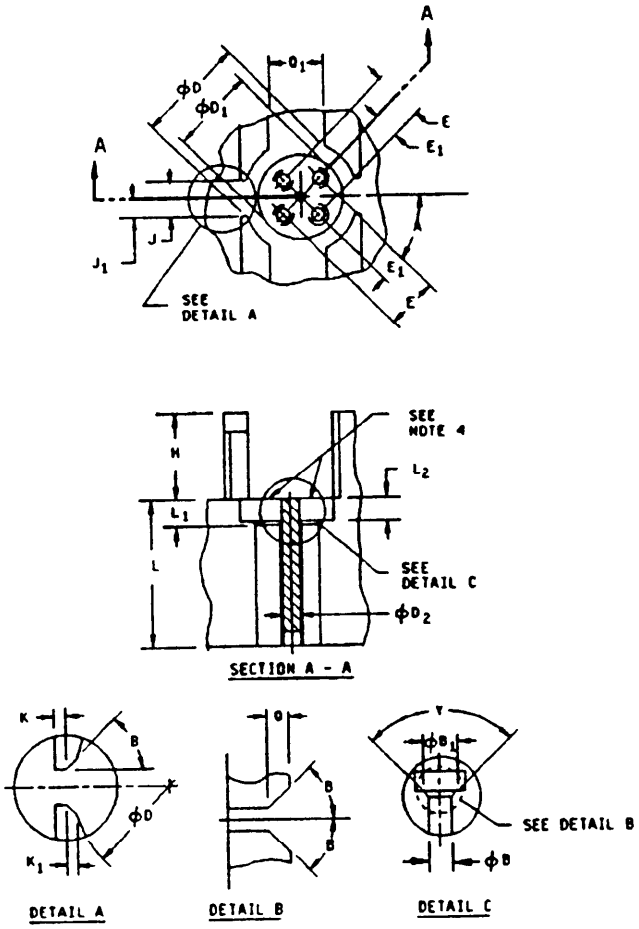
FIGURE 1. Physical dimensions (TO-18).

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Measured in the zone beyond .250 inch (6.35 mm) from the seating plane.
4. These dimensions shall be measured at points .050 inch (1.27 mm) to .250 inch (6.35 mm) from the seating plane.
5. When measured in a gauging plane .054 + .001, - .001 inch (1.37 +.03, -.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 inch (.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 2 shows the prefered measurement method.
6. The collector shall be internally connected to the case.
7. Measured from the maximum diameter of the actual device.
8. All three leads (see 3.3.1).
9. Tab centerline.

FIGURE 1. Physical dimensions (TO-18) - Continued.

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NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The following gauging procedures shall be used: The device being measured shall be inserted until its seating plane is .125 inch (3.18 mm) +.010 inch (0.254 mm) from the seating surface of the gauge. A force of 8 ±.5 oz. shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gauge. The use of a pin straightener prior to insertion in the gauge is permissible. A spacer may be used to obtain the .125 inch (3.18 mm) distance from the gauge seat prior to force application.
4. These surfaces to be parallel and in same plane within ±.001 inch (0.025 mm).
5. Four holes.
6. Pressed in.

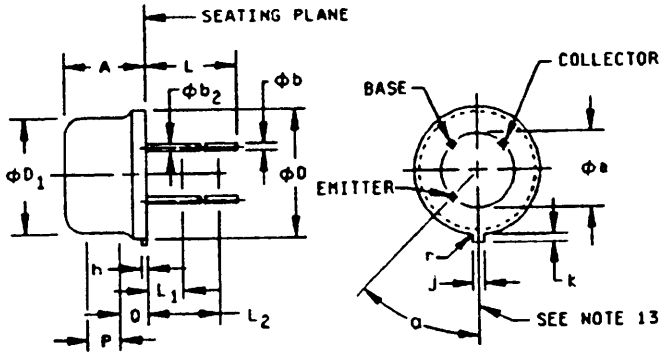
FIGURE 2. Gauge for lead and tab location.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
ϕB	.0325	.0335	.8255	.8509	5
ϕB_1	.043 Nom		1.09 Nom		5
ϕD	.2310	.2315	5.867	5.880	
ϕD_1	.159	.161	4.04	4.09	
ϕD_2	.040 Nom		1.02 Nom		6
E	.0995	.1005	2.527	2.553	
E_1	.0495	.0505	1.257	1.283	
H	.145	.155	3.68	3.94	
J	.0470	.0475	1.194	1.207	
J_1	.0235	.0245	.597	.622	
K	.009	.011	.229	.279	
K_1	.005 Nom		.127 Nom		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	
L_2	.043 Nom		1.09 Nom		
Q	.040 Nom		1.02 Nom		
Q_1	.123	.127	3.12	3.23	
A	44.90	45.10°			
B	45° Nom				
Y	90° Nom				

FIGURE 2. Gauge for lead and tab location (TO-18). Continued

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.240	.260	6.10	6.60	
Φa	.1414	Nom	3.592	Nom	7
Φb	.016	.021	.41	.53	3, 10
Φb2	.016	.019	.41	.48	4, 10
ΦD	.335	.370	8.51	9.40	
ΦD1	.305	.335	7.75	8.51	
h	.009	.125	.23	3.18	
j	.028	.034	.71	.86	
k	.029	.045	.74	1.14	9
L	See notes 10, 11, and 12				
L_1		.050		1.27	
L_2	.250		6.35		
P	.100		2.54		5
Q					6
r		.007		.18	
α	45°	T.P.	45°	T.P.	

FIGURE 3. Physical dimensions (TO-5 and TO-39).

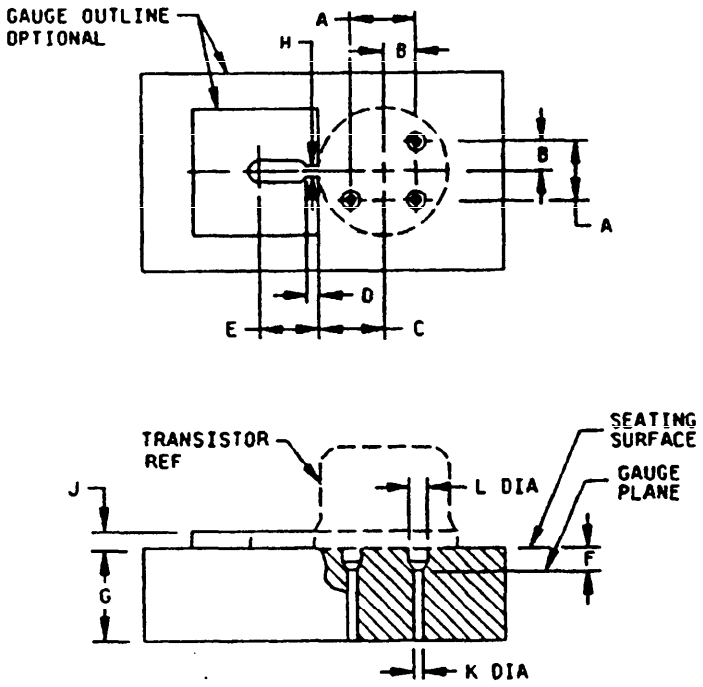
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NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Measured in the zone beyond .250 inch (6.35 mm) from the seating plane.
4. These dimensions shall be measured at points .050 inch (1.27 mm) to .250 inch (6.35 mm) from the seating plane.
5. Variations on dimension ϕD1 in this zone shall not exceed .010 inch (.25 mm).
6. Outline in this zone is not controlled.
7. When measured in a gauging plane .054 + .001, - .001 inch (1.37 +.03, -.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 inch (.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 4 shows the prefered measurement method.
8. The collector shall be internally connected to the case.Tab centerline.
9. Measured from the maximum diameter of the actual device.
10. All three leads (see 3.3.1).
11. For transistor type 2N1613, L = .500 inch (12.70 mm) minimum and .750 inch (19.05 mm) maximum.
12. For transistor type 2N1613L, L = 1.500 inch (38.10 mm) minimum and 1.750 inch (44.45 mm) maximum.
13. Tab centerline.

FIGURE 3. Physical dimensions (TO-5 and TO-39) - Continued.

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.1409	.1419	3.58	3.60
B	.0702	.0712	1.78	1.81
C	.182	.199	4.62	5.05
D	.009	.011	0.23	0.28
E	.125 Nom		3.18 Nom	
F	.054	.055	1.37	1.40
G	.372	.378	9.45	9.60
H	.035	.0355	0.89	0.90
J	.150 Nom		3.81 Nom	
K	.0325	.0335	0.83	0.85
L	.0595	.0605	1.51	1.54

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The following gauging procedure shall be used: The use of a pin straightener prior to insertion in the gauge is permissible. The device being measured shall be inserted until its seating plane is .125 ±.010 (3.18 ±0.25 mm) from the seating surface of the gauge. A spacer may be used to obtain the .125 (3.18 mm) distance from the gauge seat prior to force application. A force of 8 oz ±.50 oz shall then be applied parallel and symmetrical to the advice's cylindrical axis. When examined visually after the force application (the force need not be removed), the seating plane of the device shall be seated against the gauge.
4. The location of the tab locator, within the limits of dimension C, will be determined by the tab and flange dimension of the device being checked.

FIGURE 4. Gauge for lead and tab location.

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3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500 and paragraph 4.4.4 herein.

4.3 Screening (JANTX and JANTXV levels). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement
	JANTX and JANTXV levels only
1/	Thermal impedance (see 4.3.2)
9	Not applicable
11	h_{FE3}; I_{CBO1}
12	See 4.3.1
13	ΔI_{CBO1} = 100 percent of initial value or 5 nA dc; whichever is greater; Δh_{FE3} = ± 15 percent of initial value; subgroup 2 of table I herein.

1/ Thermal impedance shall be performed anytime after sealing provided temperature cycling is performed in accordance with MIL-S-19500, screen 3 prior to this thermal test.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_A = room ambient as defined in the general requirements of MIL-STD-750.
V_{CB} = 20 V dc.
2N718A , P_T = 500 mW dc.
2N1613, 2N1613L , P_T = 800 mW dc.

NOTE: No heat sink or forced air cooling of the devices shall be permitted.

4.3.2 Thermal impedance $Z_{\theta JX}$ measurements for screening. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit and conditions for $Z_{\theta JX}$ in screening (table II of MIL-S-19500) shall be derived by each vendor by means of process control of actual measurements which characterizes the die attach process. When three lot date codes have exhibited control, the data from these three lots will be used to establish a fixed screening limit (not to exceed the group A limit). Once a fixed limit has been established, monitor all future sealing lots using a sample from each lot.

4.3.2.1 Thermal impedance ($Z_{\theta JX}$ measurements) for initial qualification or requalification. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101 (read and record value $Z_{\theta JX}$) derived conditions, limits, and thermal response curve shall be supplied to the qualifying activity on the qualification lot prior to qualification approval.

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4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. The following test conditions shall be used for $Z_{\theta JX}$ in group A subgroup 2 inspection:

a. I_H measuring current: 10 mA.

b. I_H forward heating current: 10 A.

c. t_H heating time: 10 ms.

d. t_{MD} measurement delay time: 100 μs maximum.

The maximum limit for $Z_{\theta JX}$ in group A, subgroup 2 is $Z_{\theta JX}$(maximum) = 4.5°C/W.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, table IVb of MIL-S-19500.

Subgroup	Method	Conditions
B3	1027	T_A = room ambient as defined in the general requirements of MIL-STD-750. V_{CB} = 20 V dc. 2N718A, P_T = 500 mW dc; 2N1613, 2N1613L, P_T = 800 mW dc.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and as follows. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1 Group C inspection table V of MIL-S-19500.

Subgroup	Method	Conditions
C2	2036	Test condition E.
C6	1026	T_A = room ambient as defined in the general requirements of paragraph 4.5 of MIL-STD-750. V_{CB} = 20 V dc. 2N718A , P_T = 500 mW dc; 2N1613, 2N1613L , P_T = 800 mW dc.
C7	3131	See 4.5.2

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-S-19500, and as follows. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.4.1 Group E inspection table VII of MIL-S-19500.

Subgroup	Method	Conditions
E1	1051	Condition C, 500 cycles; 22 devices, c = 0.
E1	1071	
E2	1038	80 to 85% of rated V_R, T_A = 150°C, t = 1,000 hours; 22 devices, c = 0.
E3		Not Applicable
E4	3131	See 4.5.2, 22 devices, c = 0.
E5		Not applicable

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4.5 Method of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power applications shall be 28 mA dc (2N718A), and 47 mA dc (2N1613 and 2N1613L).

b. Collector to emitter voltage magnitude ≥ 20 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be 25°C ≤ T_R ≤ 75°C and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to case.

f. Maximum limit shall be $R_{\theta JC}$ = (2N718) 97°C/W, and (2N1613 and 2N1613L) 58°C/W.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Breakdown voltage collector-to-base	3001	Bias condition D, I_C = 100 µA dc, pulsed (see 4.5.1)	$V_{(BR)CBO}$	75		V dc
Breakdown voltage emitter-to-base	3026	Bias condition D, I_E = 100 µA dc, pulsed (see 4.5.1)	$V_{(BR)EBO}$	7		V dc
Breakdown voltage collector-emitter	3011	Bias condition D, I_C = 100 mA dc, pulsed (see 4.5.1)	$V_{(BR)CEO}$	30		V dc
Breakdown voltage collector-emitter	3011	Bias condition D, I_C = 100 mA dc, pulsed (see 4.5.1), R_{BE} = 10 ohms	$V_{(BR)CER}$	50		V dc
Collector to base cutoff current	3036	Bias condition D V_{CB} = 60 V dc	I_{CBO1}		10	nA dc
Emitter-base cutoff current	3061	Bias condition D, V_{EB} = 5 V dc	I_{EBO}		10	nA dc
Collector-emitter saturated voltage	3071	I_C = 150 mA dc, I_B = 15 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$		1.5	V dc
Base-emitter saturated voltage	3066	Test condition A, I_C = 150 mA dc, I_B = 15 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$		1.3	V dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 0.1 mA dc, pulsed (see 4.5.1)	h_{FE1}	20		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 10 mA dc, pulsed (see 4.5.1)	h_{FE2}	35		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 150 mA dc, pulsed (see 4.5.1)	h_{FE3}	40	120	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 500 mA dc, pulsed (see 4.5.1)	h_{FE4}	20		
Thermal impedance 2/	3101	See 4.5.1	$Z_{\theta JX}$		4.5	°C/W

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO2}		10	µA dc
Low temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 10 mA dc, pulsed (see 4.5.1)	h_{FE5}	20		
Subgroup 4						
Small signal short circuit forward current transfer ratio	3206	V_{CE} = 5 V dc, f = 1 kHz, I_C = 1 mA dc:	h_{fe}	30	100	
		V_{CE} = 10 V dc, f = 1 kHz, I_C = 5 mA dc	h_{fe}	35	150	
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc, I_C = 50 mA dc, f = 20 MHz	$\|h_{fe}\|$	3		
Small signal short circuit input impedance	3201	V_{CB} = 10 V dc, I_C = 5 mA dc,	h_{ib}	4	8	ohms
Small signal short circuit output admittance	3216	V_{CB} = 10 V dc, I_C = 5 mA dc,	h_{ob}	0	1.0	µohms
Small signal open circuit reverse voltage transfer ratio	3211	V_{CB} = 10 V dc, I_C = 5 mA dc,	h_{rb}		$3x10^{-4}$	
Open circuit output capacitance	3236	V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{obo}		25	pF
Pulse response:	3251	Test condition A, except test circuit and pulse requirements See figure 5 herein.	$t_{on} + t_{off}$		30	ns
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.
2/ Need not be performed in screen 13.

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TABLE II. Groups B, C, and E electrical end-point measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector to base cutoff current	3036	Bias condition D, V_{CB} = 60 V dc	I_{CBO1}		10	nA dc
2.	Collector to base cutoff current	3036	Bias condition D, V_{CB} = 60 V dc	I_{CBO1}		20	nA dc
3.	Collector to emitter voltage (saturated)	3071	I_B = 150 mA dc, I_C = 15 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$		1.5	V dc
4.	Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 150 mA dc, pulsed (see 4.5.1)	h_{FE3}	40	120	
5.	Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 150 mA dc, pulsed (see 4.5.1)	Δh_{FE3}	± 15 percent change from initial value		

1/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table III herein, steps 1, 3, and 5.

b. Subgroup 3, see table III herein, steps 2 and 4.

c. Subgroup 6, see table III herein, steps 2 and 4.

2/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroup 2, see table III herein, steps 1, 3, and 5.

b. Subgroup 3, see table III herein, steps 2 and 4.

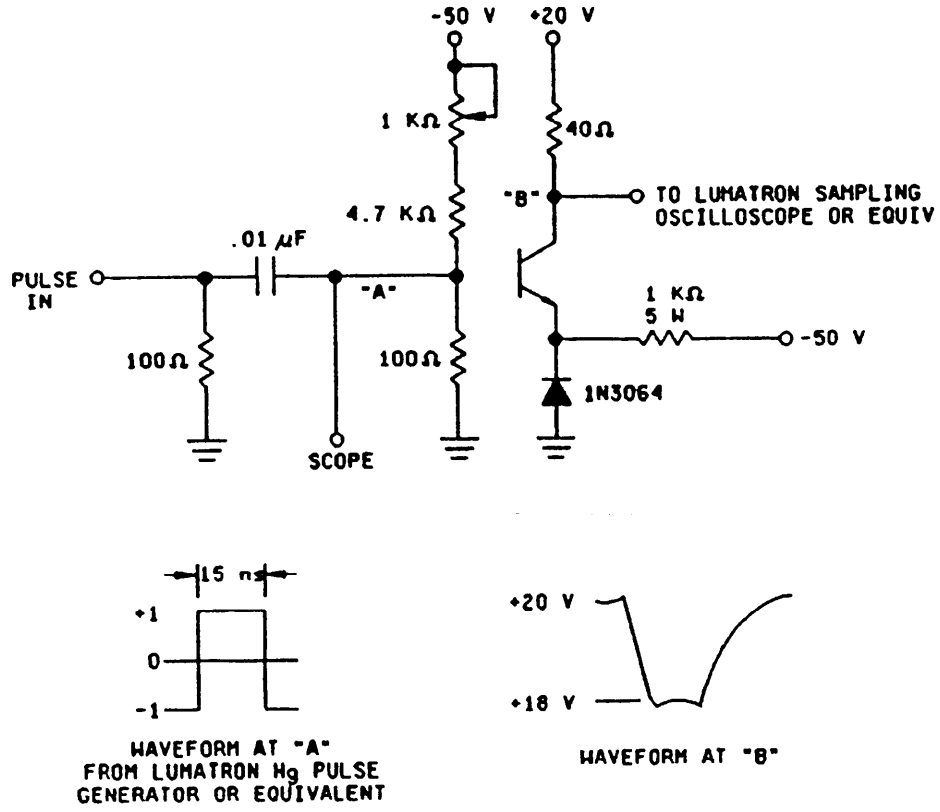
c. Subgroup 6, see table III herein, steps 2 and 4.

3/ The electrical measurements for table VII of MIL-S-19500 are as follows:

a. Subgroup 1, see table III herein, steps 1 and 2.

b. Subgroup 2, see table III herein, steps 1 and 2.

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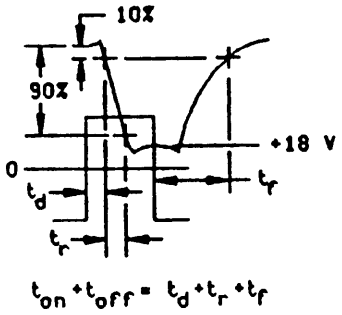


FIGURE 5. Pulse response (turn-on plus turn-off) measurement circuit and waveforms.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Type designation and quality assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, AV, MI, SM
Navy - AS, CG, MC, OS, SH
Air Force - 13, 15, 19, 85, 99
DLA - ES

Preparing activities:
Navy - EC

Agent:
DLA - ES

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/181E	2. DOCUMENT DATE (YYMMDD) 931025

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER TYPES 2N718A, 2N1613, AND 2N1613L, JANTX AND JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, first, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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